ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 2 [10 marks]

Q2(a) [2 marks]

The state model is $\dot{x}_1 = x_2$, $\dot{x}_2 = (\tau - 0.05x_2 - 2.4525 \sin x_1)/0.02$. Equilibrium requires $\tau_0 = 2.4525 \sin \theta_e$, so τ_0 must be between -2.4525 and +2.4525 N m.

Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60^\circ)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$
$$B = [[0], [50]]$$

Q2(c) [4 marks]

At steady state the derivative terms vanish. The proportional torque is Kp(θd-θ), which must balance mgℓ sin θ.

Q2(d) [2 marks]

The gravity term changes the equilibrium, so the steady-state error is still nonzero. The poles should change.

Question 3 [10 marks]

Q3(a) [2 marks]

Finite differencing amplifies encoder noise, making the velocity estimate noisy.

Q3(b) [2 marks]

Tustin uses s = (2/T)(z-1)/(z+1).

Q3(c) [4 marks]

$$L = [18, 64]^T$$

Q3(d) [2 marks]

An observer can estimate velocity from position measurements. A filter can be safer because it smooths noise.

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

The actual feedforward contribution is 0.75x8=6.

Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.